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C:\Users\admin>wsl
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adminayushi@DESKTOP-M312757:/mnt/c/Users/admin$ cat client.c
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#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

//Server code
#define PORT 8081
#define BUFFER_SIZE 1024

int main(){
    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);

    char buffer[BUFFER_SIZE];
    char message[BUFFER_SIZE];

    //Socket creation
    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    if (server_fd < 0){
        perror("Socket creation failed");
        exit(1);
    }
    //configure address
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);
    //Bind
    if (bind(server_fd, (struct sockaddr *)&address, sizeof(address))<0){
        perror("Bind failed");
        exit(1);
    }
    //listen for connections
    if (listen (server_fd,5)<0) {
        perror ("listen failed");
        exit(1);
    }
    printf("Server listening on the port %d \n", PORT);

    // Accept connection

    new_socket = accept(server_fd, (struct sockaddr *)& address, (socklen_t *) & addrlen);

    if(new_socket<0){
        perror("Accept failed");
        exit(1);
    }
}
```

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}

printf("client communicated \n");

//communication loop

while(1){
    read(new_socket,buffer,1024);
    printf("client:%s",buffer);
    if(strcmp(buffer,"bye\n") ==0){
        break;
    }
    printf("server:");
    fgets(message,sizeof(message),stdin);
    send(new_socket,message,strlen(message),0);
    if(strcmp(buffer,"bye\n") ==0){
        break;
    }
}

close(new_socket);
close(server_fd);
return 0;
}

adminayushi@DESKTOP-M312757:/mnt/c/Users/admin$ cat server.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 8081
#define BUFFER_SIZE 1024

//CLIENT CODE
int main(){
    int sock;
    struct sockaddr_in serv_addr;

    char buffer[BUFFER_SIZE];
    char message[BUFFER_SIZE];
    // SOCKET creation
    sock = socket(AF_INET, SOCK_STREAM,0);

    if (sock<0){
        perror("socket creation failed");
        exit(1);
    }

    //configure server address
```

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    perror("socket creation failed");
    exit(1);
}

//configure server address
serv_addr.sin_family = AF_INET;
serv_addr.sin_port = htons(PORT);
inet_pton(AF_INET, "127.0.0.1",&serv_addr.sin_addr);
//connect to server
if (connect(sock,(struct sockaddr *)& serv_addr,sizeof(serv_addr))<0){
    perror("connection failed");
    exit(1);
}

printf("connected to server");
//loop

while(1){
    printf("client:");
    fgets(message, sizeof(message),stdin);
    send(sock, message,strlen(message),0);
    if(strcmp(buffer,"bye\n") ==0){
        break;
    }
    read(sock,buffer,1024);
    printf("Server:%s",buffer);
    if(strcmp(buffer,"bye\n") ==0){
        break;
    }
}
close(sock);
return 0;
}
adminayushi@DESKTOP-M312757:/mnt/c/Users/admin$
```